

QRAFT TECHNOLOGIES, MULTI ASSET TEAM, 2025

Graph-Based Dynamic Clustering of QIS Strategies

Idea overview — letting the correlation structure of hundreds of quantitative investment strategies define the portfolio, instead of static factor labels.

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Static factor labels miss how strategies co-move

Why a QIS portfolio built on pre-defined factor buckets ends up less diversified than it looks



Labels are static

Each QIS is filed once under a factor name — momentum, carry, value — and never moves, even when its behaviour does.



Correlations change with regimes

Dependencies among strategies are time-varying. A grouping that was orthogonal in 2019 can be crowded in 2022.



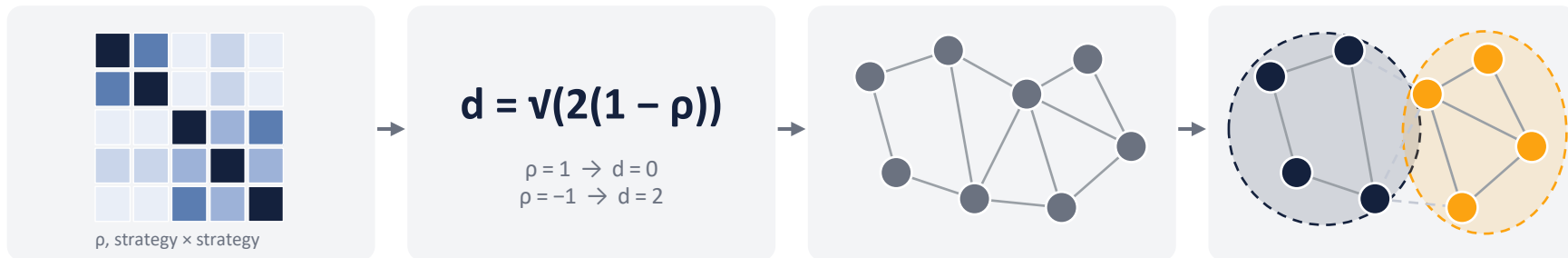
Hundreds of overlapping strategies

Multi-bank universes (BofA, Macquarie, Morgan Stanley via Premialab) contain many near-duplicates under different names.

Result: a portfolio that is diversified on paper but concentrates risk in stress — exactly when diversification is needed.

Idea: let the data define the clusters

An unsupervised, deterministic pipeline from daily returns to a hierarchy of strategy communities



1 Correlation

Daily returns over a 3-year rolling window give the empirical correlation matrix ρ between every pair of strategies.

2 Distance

$d = \sqrt{2(1 - \rho)}$ turns correlation into a proper distance: identical strategies sit at 0, opposite ones at 2.

3 PMFG

The Planar Maximally Filtered Graph keeps only the strongest links that fit on a plane — more information than a spanning tree, far less noise than the full matrix.

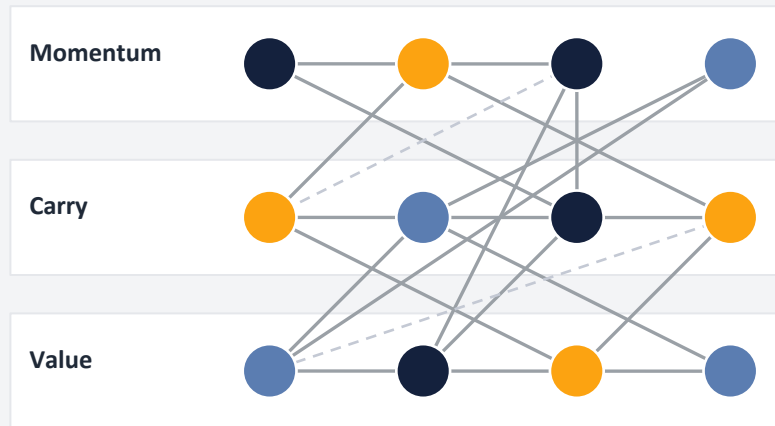
4 DBHT

The Directed Bubble Hierarchical Tree reads communities and their hierarchy from the graph's 3-cliques. No number of clusters to choose.

Same strategies, two ways of grouping them

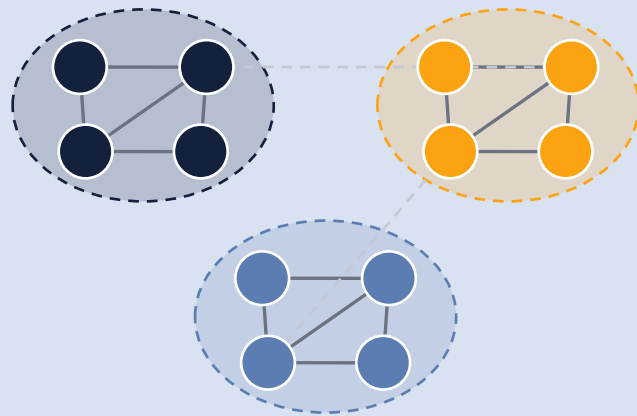
Twelve strategies; edges are their strongest correlations. Colour = the community the data actually forms.

Grouped by provider factor label



Most strong links cross the label boundaries: the buckets are not the real structure.

Grouped by DBHT cluster

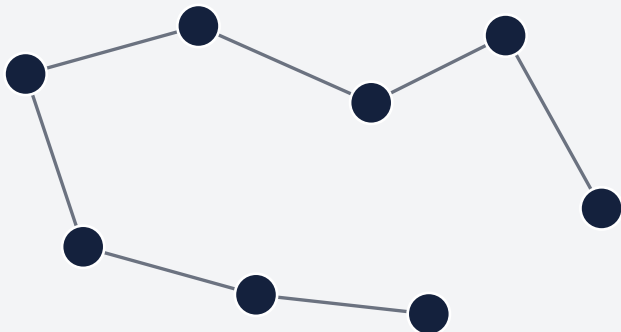


Same edges, regrouped: strong links stay inside a cluster, only weak ones cross.

Why a filtered graph: spanning tree vs PMFG

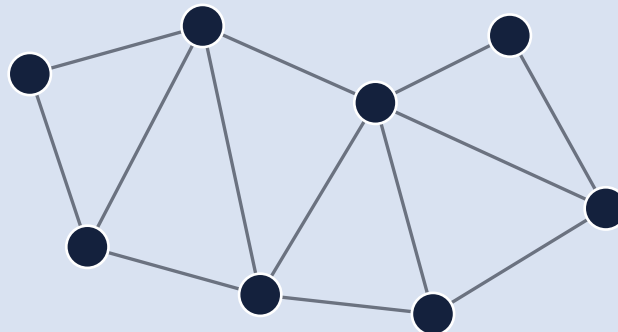
Same eight strategies, same distances. A tree keeps $N - 1$ links; the PMFG keeps up to $3(N - 2)$ while staying planar.

Minimum spanning tree: 7 links



One path between any two strategies; no triangles, so no local structure to read.

PMFG: 13 links here, no crossings

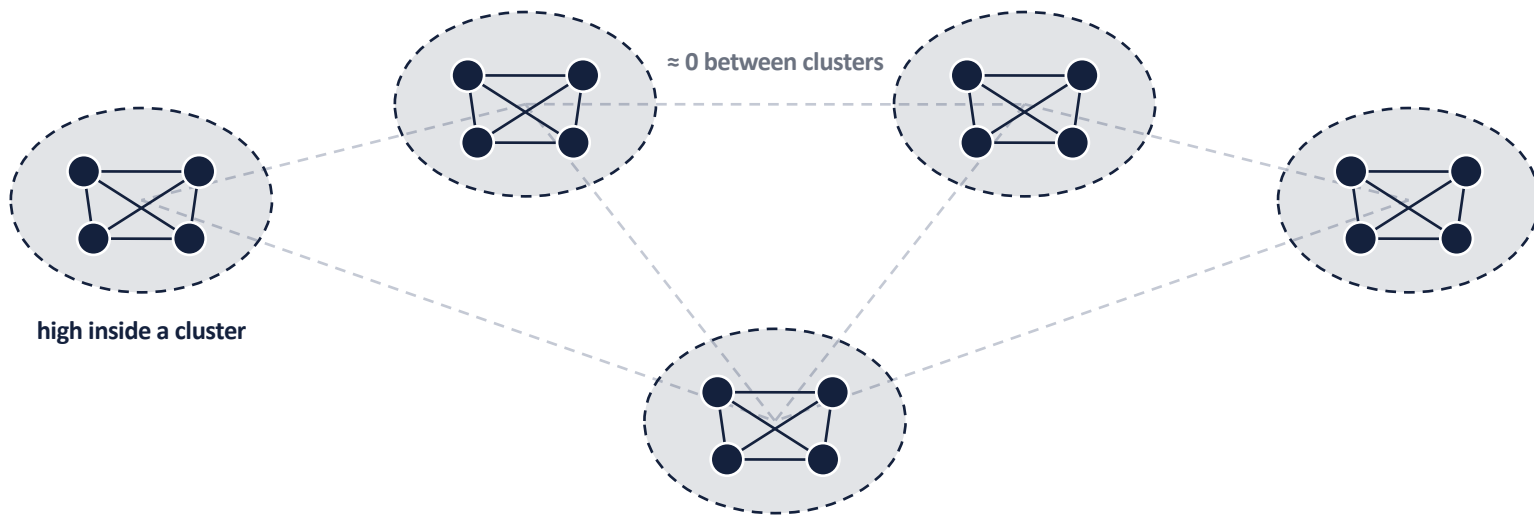


Keeps the triangles (3-cliques) that DBHT reads as clusters and hierarchy.

- PMFG retains more information than a spanning tree with looser topological constraints; DBHT then finds the number of clusters from the data — no k to choose.
- Re-estimated every quarter and fully deterministic: the same inputs always give the same clusters.

Evidence: DBHT clusters are more orthogonal

3-year rolling correlations, DBHT clusters vs. static factor groups, on the Premialab QIS universe (source: Premialab, Qraft Technologies)



Inter-group correlation ↓

Equal-weighted cluster indices stay close to zero correlation, well below static factor groups

Intra-group correlation ↑

Members of a DBHT cluster are consistently more correlated than members of a factor group

From clusters to a portfolio

Quarterly, rule-based, equal-weighted — 42 candidate sub-portfolios, one survivor

- 1 Cluster** 1st business day of each quarter: 5 DBHT clusters over the QIS universe
- 2 Generate** 2 representative QIS per cluster, equal-weighted → 42 sub-portfolios
- 3 Screen** CAGR > 3%, Sharpe > 0.8, volatility < 6%, max drawdown > -6%
- 4 Hold** Equal weights: robust to estimation error, keeps the diversification the clusters found

	3M	6M	1Y	2Y	3Y	4Y	5Y
CAGR high					★		
CAGR low							
Sharpe high							
Sharpe low							
Skew high							
Skew low							

42 = 3 metrics × high / low × 7 lookbacks. ★ highest CAGR with a 3-year lookback — the configuration that passed every screen and was carried forward.

Simulation result, Jan 2019 – Sep 2024

10-QIS equal-weighted portfolio from 5 clusters, highest-CAGR selection, 3-year lookback, quarterly rebalance, 40 bp costs

5.24%

CAGR

1.04

Sharpe ratio

-5.40%

Max drawdown

4.25%

Annualised volatility

34.05%

Cumulative return

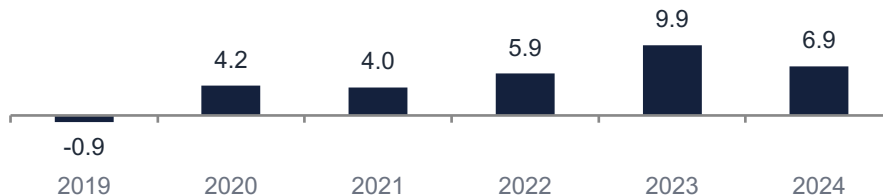
1.40

Sortino ratio

CUMULATIVE RETURN, NORMALISED AT 100 ON 1 JAN 2019



YEARLY RETURN (%)



Universe: BofA, Macquarie, Morgan Stanley QIS via Premialab. Curve redrawn from Figure 3 of the Qraft white paper; statistics are from the original daily series.

Takeaways

Correlation structure, not factor labels, should decide what diversifies what — and it should be re-estimated as regimes change.

PMFG + DBHT gives that structure unsupervised and deterministically, with an interpretable hierarchy.

A simple rule-based, equal-weighted selection on top of the clusters delivered Sharpe 1.04 with a -5.4% max drawdown over 2019–2024 after costs.

Validated in proof-of-concept projects with major global financial institutions.

Python, SQL, NetworkX, PMFG and DBHT, scikit-learn, PyTorch

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